Real Options with Poisson Exercise Opportunities

An Elliptic Model and the POST Iterative Method

Abstract

We formulate a perpetual real-option valuation problem with two correlated diffusion factors and exercise restricted to the arrival times of an independent Poisson process. The resulting equation is local and semilinear elliptic. We specify its natural state space, the conditions selecting the financial solution, a bounded-domain Dirichlet approximation, and the Poisson optional stopping times (POST) iteration. An abandonment example provides explicit payoffs, parameters, and computational boundary data. The general valuation equation and POST recursion follow Lange, Ralph, and Støre [1]; the abandonment specialization, boundary closure, and estimates below are derived here to give a self-contained numerical problem.

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1 Economic setting and assumptions

1.1 Two stochastic factors

Let X_t denote the project's operating-cost rate and Y_t its revenue rate. Under a specified valuation measure, assume

$$dX_t = \mu_X X_t\,dt + \sigma_X X_t\,dW_t^X, \tag{1.1}$$

$$dY_t = \mu_Y Y_t\,dt + \sigma_Y Y_t\,dW_t^Y, \qquad d\langle W^X, W^Y\rangle_t = \rho\,dt. \tag{1.2}$$

Here $\sigma_X, \sigma_Y > 0$ and $|\rho| < 1$. The rate $r > 0$ discounts cash flows. In a replicable risk-neutral model, the drifts and r are risk-neutral drifts and the risk-free rate. Alternatively, these equations can describe a decision model under the physical measure with a specified risk-adjusted discount rate; these interpretations must not be mixed in a calibration.

From an initial state (x, y) with $x, y > 0$, both factors remain positive at every finite time. The natural open state space is therefore

$$D = (0, \infty)^2. \tag{1.3}$$

We assume time-independent running cash flow $f : D \to \mathbb{R}$ and stopping payoff $G : D \to \mathbb{R}$. The running cash flow is paid while the project operates; G is received once, when operation ends.

For a simple sufficient integrability assumption, put

$$w(x, y) = 1 + x + y, \qquad \alpha = \max\{0, \mu_X, \mu_Y\}, \qquad r > \alpha, \qquad |f(x, y)| + |G(x, y)| \le Cw(x, y). \tag{1.4}$$

The constant 1 in w is understood after choosing reference units for the factors. Continuity of f and G is assumed. The growth condition is sufficient, rather than a necessary condition for every model in [1].

1.2 Permitted exercise times and value functions

Let N_t be a Poisson process with rate $\lambda > 0$, independent of the Brownian motions. Its arrival times are $0 < T_1 < T_2 < \cdots$. Let $\mathcal{T}_\lambda$ contain stopping times for the joint filtration that take values in $\{T_1, T_2, \ldots\} \cup \{\infty\}$. At an arrival, the owner may stop or continue; before the first arrival and between arrivals the owner must continue.

Define the value at a time when no immediate exercise opportunity is available:

$$V_\lambda(x, y) = \sup_{\tau \in \mathcal{T}_\lambda} \mathbb{E}_{x,y}\left[\int_0^\tau e^{-rt} f(X_t, Y_t)\,dt + \mathbf{1}_{\{\tau < \infty\}} e^{-r\tau} G(X_\tau, Y_\tau)\right]. \tag{1.5}$$

For $\tau = \infty$, the terminal payoff is omitted. The assumptions in (1.4) make these expected absolute payoffs finite. At an exercise opportunity, the value is instead

$$U_\lambda(x, y) = \max\{G(x, y), V_\lambda(x, y)\}. \tag{1.6}$$

An optimal rule stops at the first arrival with $G(X_{T_n}, Y_{T_n}) \ge V_\lambda(X_{T_n}, Y_{T_n})$, whenever this rule realizes the value; either action is optimal at equality.

2 The stationary elliptic equation

2.1 Generator and Bellman equation

The diffusion generator is

$$\begin{aligned}\mathcal{L}v = {} & \tfrac{1}{2}\sigma_X^2 x^2 v_{xx} + \rho\sigma_X\sigma_Y xy v_{xy} + \tfrac{1}{2}\sigma_Y^2 y^2 v_{yy} \\ & + \mu_X x v_x + \mu_Y y v_y.\end{aligned} \tag{2.1}$$

With probability $\lambda h + o(h)$ an opportunity arrives in a short interval of length h; its extra value is $U_\lambda - V_\lambda$. Combining this with the running cash flow and the generator yields

$$rV_\lambda = \mathcal{L}V_\lambda + f + \lambda\big(U_\lambda - V_\lambda\big). \tag{2.2}$$

Consequently the governing equation, valid throughout D, is

$$\boxed{\mathcal{L}V_\lambda - rV_\lambda + f + \lambda\,[G - V_\lambda]^+ = 0 \quad \text{in } D.} \tag{2.3}$$

Equivalently,

$$\boxed{(r + \lambda - \mathcal{L})V_\lambda = f + \lambda \max\{G, V_\lambda\}.} \tag{2.4}$$

These equalities are classical wherever the solution has the requisite derivatives, and admit a viscosity interpretation more generally.

The stopping set at arrival times is $S_\lambda = \{G \geq V_\lambda\}$, and the continuation set is $C_\lambda = \{G < V_\lambda\}$. The interface $G = V_\lambda$ is found after solving (2.3); it is not an additional prescribed boundary. In particular, *do not impose $V_\lambda \geq G$ throughout D*. This value is measured between opportunities and may lie below G. Also, $V_\lambda = G$ throughout S_λ and the usual American-option smooth-pasting conditions are not boundary conditions for this finite-λ model. The state factors have continuous paths; the Poisson exercise clock introduces the local reaction term in (2.3), not a jump integral.

2.2 Ellipticity and logarithmic coordinates

Write the principal part as $\frac{1}{2}\operatorname{tr}(aD^2v)$, with

$$a(x,y) = \begin{pmatrix} \sigma_X^2 x^2 & \rho\sigma_X\sigma_Y xy \\ \rho\sigma_X\sigma_Y xy & \sigma_Y^2 y^2 \end{pmatrix}. \tag{2.5}$$

Its determinant is $\sigma_X^2\sigma_Y^2 x^2 y^2(1-\rho^2) > 0$. Thus a is positive definite at every point of D. It is not uniformly positive definite on the whole original domain, since coefficients vanish as a factor approaches zero. On a compact rectangle inside D it is uniformly positive definite.

For reference levels $x_*, y_* > 0$, set

$$\xi = \log(x/x_*), \quad \eta = \log(y/y_*), \quad v(\xi,\eta) = V_\lambda(x_* e^\xi, y_* e^\eta). \tag{2.6}$$

The transformed natural domain is $\mathbb{R}^2$, and

$$\widetilde{\mathcal{L}}v = \tfrac{1}{2}\sigma_X^2 v_{\xi\xi} + \rho\sigma_X\sigma_Y v_{\xi\eta} + \tfrac{1}{2}\sigma_Y^2 v_{\eta\eta} + \beta_X v_\xi + \beta_Y v_\eta, \quad \beta_i = \mu_i - \tfrac{1}{2}\sigma_i^2. \tag{2.7}$$

The principal matrix is the constant positive-definite matrix

$$A = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}. \tag{2.8}$$

For every $p \in \mathbb{R}^2$, $p^\top A p \geq \kappa|p|^2$ with $\kappa = \lambda_{\min}(A) > 0$. The log-coordinate problem is therefore *uniformly elliptic*:

$$\widetilde{\mathcal{L}}v - rv + \widetilde{f} + \lambda\left[\widetilde{G} - v\right]^+ = 0 \quad \text{in } \mathbb{R}^2, \qquad (\widetilde{f},\widetilde{G})(\xi,\eta) = (f,G)(x_* e^\xi, y_* e^\eta). \tag{2.9}$$

Uniform ellipticity does not imply a bounded solution on $\mathbb{R}^2$; the growth condition still matters.

3 Natural conditions and computational boundary conditions

3.1 The unbounded-domain financial problem

There is no finite artificial boundary in (1.3). The axes cannot be reached in finite time from D, so arbitrary Dirichlet or reflecting conditions on $x = 0$ and $y = 0$ should not be appended. The financial solution is selected in the class

$$|V_\lambda(x,y)| \le C_V w(x,y), \qquad \lim_{t\to\infty} \mathbb{E}_{x,y}[e^{-rt}|V_\lambda(X_t,Y_t)|] = 0. \tag{3.1}$$

The second condition follows from the first and $r > \alpha$ because $\mathbb{E}_{x,y} w(X_t,Y_t) \le e^{\alpha t} w(x,y)$. In log coordinates, these are growth and transversality conditions as $|\xi| + |\eta| \to \infty$, not zero values at infinity. If one extends the model to an initial state on an axis, the corresponding factor stays zero and the valuation problem reduces to a one-factor problem.

3.2 A bounded Dirichlet approximation

For computation, select

$$D_B = (x_{\min}, x_{\max}) \times (y_{\min}, y_{\max}), \quad 0 < x_{\min} < x_{\max}, \quad 0 < y_{\min} < y_{\max}. \tag{3.2}$$

Let B be a specified continuous boundary-value approximation. The bounded problem is precisely

$$\boxed{\begin{cases} (r + \lambda - \mathcal{L})V_B = f + \lambda \max\{G, V_B\}, & \text{in } D_B, \\ V_B = B, & \text{on } \partial D_B. \end{cases}} \tag{3.3}$$

Equivalently, solve the transformed PDE on

$$\Omega = (\log(x_{\min}/x_*), \log(x_{\max}/x_*)) \times (\log(y_{\min}/y_*), \log(y_{\max}/y_*)), \quad v_B = \widetilde{B} \text{ on } \partial\Omega. \tag{3.4}$$

The boundary ∂D_B is computational and is unrelated to ∂C_λ. Taking $B = V_\lambda|_{\partial D_B}$ would give the exact restriction of the financial solution, but this trace is unknown. Section 5 gives an explicit practical approximation.

The Dirichlet problem has a useful exact interpretation: let ζ_B be the first exit time from D_B. At exit, the project is terminated with payoff $B(X_{\zeta_B}, Y_{\zeta_B})$; before exit, stopping is permitted only at the Poisson times. Its value is

$$\begin{aligned} V_B(x,y) = \sup_{\tau\in\mathcal{T}_\lambda} \mathbb{E}_{x,y}\Bigg[& \int_0^{\tau\wedge\zeta_B} e^{-rt} f(X_t,Y_t)\,dt \\ & + \mathbf{1}_{\{\tau<\zeta_B\}} e^{-r\tau} G(X_\tau,Y_\tau) \\ & + \mathbf{1}_{\{\zeta_B\le\tau\}} e^{-r\zeta_B} B(X_{\zeta_B},Y_{\zeta_B}) \Bigg]. \end{aligned} \tag{3.5}$$

This interpretation makes clear what an approximate boundary closure changes. The reaction function $rs - \lambda(G - s)^+$ is strictly increasing in s, with slope either r or $r + \lambda$. Together with ellipticity, this gives the comparison principle and uniqueness for the bounded Dirichlet problem. For two closures B_1, B_2 with the same interior data, comparison gives

$$\|V_{B_1} - V_{B_2}\|_{\infty,D_B} \le \|B_1 - B_2\|_{\infty,\partial D_B}. \tag{3.6}$$

This bound is useful only when the boundary discrepancy is known or bounded.

4 POST: a sequence of linear elliptic solves

4.1 Original recursion on the natural state space

Set $s = r + \lambda$. The POST sequence is initialized by

$$(s - \mathcal{L})V^{(1)} = f + \lambda G, \tag{4.1}$$

with the growth condition in (3.1). For $k \geq 1$, solve

$$\boxed{(s - \mathcal{L})V^{(k+1)} = f + \lambda \max\{G, V^{(k)}\}.} \tag{4.2}$$

The diffusion operator is identical at every iteration. The first iterate forces stopping at T_1; the kth iterate permits rejecting up to $k-1$ opportunities and forces stopping at T_k. The feasible policies are nested, so

$$V^{(1)} \leq V^{(2)} \leq \cdots \leq V_\lambda. \tag{4.3}$$

No nonnegativity assumption on the first iterate is required.

To derive the recursion directly, define the diffusion resolvent

$$R_s h(x, y) = \mathbb{E}_{x,y} \int_0^\infty e^{-st} h(X_t, Y_t)\, dt. \tag{4.4}$$

Conditioning on the independent exponential first arrival gives

$$V_\lambda = R_s\big(f + \lambda \max\{G, V_\lambda\}\big), \quad V^{(1)} = R_s(f + \lambda G). \tag{4.5}$$

The differential identity $(s - \mathcal{L})R_s h = h$ recovers the elliptic solves.

4.2 Convergence under the stated growth assumptions

Use $\|u\|_w = \sup_D |u|/w$. Because $\mathbb{E}w(X_t, Y_t) \leq e^{\alpha t} w(x, y)$,

$$\|R_s h\|_w \leq \frac{\|h\|_w}{s - \alpha}. \tag{4.6}$$

Since $|\max(G, u) - \max(G, v)| \leq |u - v|$, the fixed-point map $\mathcal{T}u = R_s(f + \lambda \max(G, u))$ satisfies

$$\|\mathcal{T}u - \mathcal{T}v\|_w \leq q_w \|u - v\|_w, \qquad q_w = \frac{\lambda}{r + \lambda - \alpha} < 1. \tag{4.7}$$

Thus it has a unique fixed point in this growth class, and the POST iterates converge to it. A computable iteration-error estimate is

$$\left\|V_\lambda - V^{(k)}\right\|_w \leq \frac{q_w}{1 - q_w} \left\|V^{(k)} - V^{(k-1)}\right\|_w, \qquad k \geq 2. \tag{4.8}$$

This is a sufficient weighted-norm estimate derived here. It should not be confused with the pointwise geometric bound in [1], which uses additional technical conditions and the factor $\lambda/(r + \lambda)$.

4.3 POST for the bounded BVP

Keep the same boundary values at every iteration:

$$(s - \mathcal{L})V_B^{(1)} = f + \lambda G \quad \text{in } D_B, \qquad V_B^{(1)} = B \quad \text{on } \partial D_B, \tag{4.9}$$

$$(s - \mathcal{L})V_B^{(k+1)} = f + \lambda \max\{G, V_B^{(k)}\} \quad \text{in } D_B, \qquad V_B^{(k+1)} = B \quad \text{on } \partial D_B. \tag{4.10}$$

The maximum principle gives nondecreasing iterates. Differences of successive solves have homogeneous boundary values. The same principle gives an unweighted contraction factor

$$q_B = \frac{\lambda}{r+\lambda} < 1, \qquad \left\| V_B - V_B^{(k)} \right\|_\infty \le \frac{q_B}{1-q_B} \left\| V_B^{(k)} - V_B^{(k-1)} \right\|_\infty . \tag{4.11}$$

For a desired iteration error ε, stop when the last increment is at most $(1-q_B)\varepsilon/q_B$. This certifies only iteration error for exact elliptic solves, not spatial discretization or boundary-truncation error. Increasing λ makes exercise more frequent but drives q_B toward one, so the basic POST iteration becomes slower.

5 A fully specified abandonment example

5.1 Payoffs and an explicit lower boundary closure

Consider a project that can be abandoned with zero salvage value:

$$f(x,y) = y - x - c_0, \qquad G(x,y) = 0, \qquad c_0 \ge 0. \tag{5.1}$$

The running cash flow can be negative. The finite-λ value before an opportunity can also be negative, because losses must be paid until an opportunity arrives.

Stopping at the first opportunity gives the explicit feasible-policy value

$$V^{(1)}(x,y) = \frac{y}{r+\lambda-\mu_Y} - \frac{x}{r+\lambda-\mu_X} - \frac{c_0}{r+\lambda}. \tag{5.2}$$

Indeed $(r+\lambda-\mathcal{L})V^{(1)} = y - x - c_0$. Continuing forever and bounding positive revenues give useful independent checks:

$$\max\left\{ V^{(1)}, \frac{y}{r-\mu_Y} - \frac{x}{r-\mu_X} - \frac{c_0}{r} \right\} \le V_\lambda(x,y) \le \frac{y}{r-\mu_Y}. \tag{5.3}$$

The lower bound uses two admissible policies; the upper bound discards all costs and allows revenues to continue forever. Neither asserts $V_\lambda \ge 0$. As a limiting check at the absorbing initial state $(0,0)$, $V_\lambda(0,0) = -c_0/(r+\lambda)$.

For an explicit bounded BVP, choose the Dirichlet closure

$$\boxed{B(x,y) = V^{(1)}(x,y) \quad \text{on } \partial D_B.} \tag{5.4}$$

This is a stated approximation, not an exact boundary condition from the original article. At computational exit it assigns the value of the policy that subsequently abandons at its first opportunity. Since $B \le V_\lambda$ on the boundary, comparison yields

$$V^{(1)} \le V_B \le V_\lambda \quad \text{on } D_B. \tag{5.5}$$

The bounded first POST iterate is exactly (5.2). Under this closure, the second and later iterations are

$$\begin{cases} (r+\lambda-\mathcal{L})V_B^{(k+1)} = y - x - c_0 + \lambda \max\{0, V_B^{(k)}\}, & \text{in } D_B, \\ V_B^{(k+1)} = \dfrac{y}{r+\lambda-\mu_Y} - \dfrac{x}{r+\lambda-\mu_X} - \dfrac{c_0}{r+\lambda}, & \text{on } \partial D_B. \end{cases} \tag{5.6}$$

Expanding the box with this closure enlarges the available policy class, so the exact bounded values are nondecreasing on their common interior. In computations, domain expansion should be assessed in a fixed region of financial interest, separately from mesh refinement.

5.2 Illustrative numerical specification

The following values are a reproducible starting example, not an empirical calibration or the original paper's renewal case study. Time is in years; x, y, c_0 are monetary cash-flow rates.

Quantity	Value	Interpretation
r	0.08	discount rate
μ_X, μ_Y	0.02, 0.01	cost and revenue growth rates
σ_X, σ_Y	0.20, 0.25	diffusion volatilities
ρ	0.30	Brownian correlation
λ	2	two opportunities per year on average
c_0	0.10	fixed operating-cost rate
x_*, y_*	1, 1	reference factor levels
D_B	$(0.05, 5) \times (0.05, 5)$	initial computational box
(x_0, y_0)	$(1, 1.2)$	reporting state

Here $r > \alpha = 0.02$, $\beta_X = 0$, and $\beta_Y = -0.02125$. On $\Omega = (\log 0.05, \log 5)^2$, the concrete log-coordinate BVP is

$$\begin{cases} -0.020\, v_{\xi\xi} - 0.015\, v_{\xi\eta} - 0.03125\, v_{\eta\eta} \\ \quad + 0.02125\, v_\eta + 0.08 v - 2\max\{-v, 0\} = e^\eta - e^\xi - 0.10, & \text{in } \Omega, \\ v = \dfrac{e^\eta}{2.07} - \dfrac{e^\xi}{2.06} - \dfrac{0.10}{2.08}, & \text{on } \partial\Omega. \end{cases} \tag{5.7}$$

The first-iterate formula provides a direct check on coefficient signs and Dirichlet implementation before solving the nonlinear problem.

6 Discrete implementation and verification

Let L_h discretize (2.7) on the interior grid; boundary contributions from the fixed Dirichlet data form a vector d_B. With $M_h = (r + \lambda)I - L_h$, the algebraic POST solves are

$$M_h \mathbf{v}^{(1)} = \mathbf{f} + \lambda \mathbf{G} + d_B, \tag{6.1}$$

$$M_h \mathbf{v}^{(k+1)} = \mathbf{f} + \lambda \max\{\mathbf{G}, \mathbf{v}^{(k)}\} + d_B. \tag{6.2}$$

The maximum is componentwise. Assemble M_h once and reuse its factorization or preconditioner. Suitable linear-solver accuracy must be included in any iteration-error budget.

For discrete monotonicity, choose a generator discretization with nonnegative off-diagonal entries and nonpositive row sums, including boundary elimination. Then M_h is a nonsingular M-matrix with $M_h^{-1} \geq 0$ and $\left\|M_h^{-1}\right\|_\infty \leq 1/(r+\lambda)$. The monotonicity and bound (4.11) then carry over to the algebraic fixed point. Positive definiteness of the continuous diffusion matrix alone does not ensure this property for a particular stencil. In particular, a standard centered mixed-derivative stencil can introduce off-diagonal entries of the wrong sign; a monotone directional or wider stencil may be needed when $\rho \neq 0$.

A practical computation proceeds as follows:

1. Specify the payoff data, log box, and fixed boundary closure. Assemble M_h and d_B.

2. Solve the first-arrival problem; for the abandonment example, compare with (5.2) as the grid is refined.

3. Repeatedly solve (6.2). Record the last increment, the nonlinear residual, and whether the iterates increase componentwise.

4. Stop using (4.11) if the discrete comparison conditions and linear-solve accuracy support that bound.

5. Extract the arrival-time stopping set $\{G \geq v_h\}$ and report values in a fixed interior region. Refine the mesh and expand the box independently to separate iteration, discretization, and boundary effects.

For reference, the interior residual is

$$\mathbf{R}_h = M_h \mathbf{v} - \mathbf{f} - \lambda \max\{\mathbf{G}, \mathbf{v}\} - d_B. \tag{6.3}$$

Do not replace negative iterates by zero in the abandonment model: only the arrival-time choice $\max\{0, \mathbf{v}\}$ is clipped.

7 Relation to renewal and continuous exercise

The article's main numerical case study is repeated project renewal rather than one-time abandonment. There $f(x, y) = y - x$ and renewing at an opportunity resets the factors to (x_I, y_I) at cost K. Its stopping gain is endogenous: $G[V] = V(x_I, y_I) - K$. The equation becomes

$$\mathcal{L}V - rV + y - x + \lambda \left[V(x_I, y_I) - K - V(x, y)\right]^+ = 0. \tag{7.1}$$

Its POST analogue updates both the continuation and renewal values:

$$(r + \lambda - \mathcal{L})V^{(k+1)} = y - x + \lambda \max\{V^{(k)}(x_I, y_I) - K, V^{(k)}(x, y)\}. \tag{7.2}$$

Although the diffusion part is elliptic, the reset evaluation couples different points. The preceding bounded-boundary specification is for the fixed-G stopping model. A renewal computation must specify its own boundary closure, interpolation at the reset point, and an initialization justified for that model; the financial interpretation of the fixed-G first-arrival iterate must not be transferred unchanged.

Finally, under conditions ensuring convergence as $\lambda \to \infty$, the finite-opportunity model approaches the continuously exercisable problem

$$\max\{\mathcal{L}V_\infty - rV_\infty + f,\ G - V_\infty\} = 0. \tag{7.3}$$

The obstacle $V_\infty \geq G$ belongs to this limiting model. For the finite-λ problem described here, the correct equation remains the semilinear elliptic equation (2.3) on the entire state space.

References

[1] R.-J. Lange, D. Ralph, and K. Støre, *Real-Option Valuation in Multiple Dimensions Using Poisson Optional Stopping Times*, Journal of Financial and Quantitative Analysis, 55(2), 653–677, 2020. doi:10.1017/S0022109019000048. Sections II–III give the valuation equations and POST recursions; Appendices B–D discuss convergence and discretization. Open accepted manuscript.